The Costs of Production

Economic cost → production → short-run costs → two cost relationships → the long run

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1. Economic cost versus accounting cost

> **DEFINITION Explicit, implicit, and economic costs**
>
> An **explicit cost** is a payment the firm makes, such as wages, rent, or materials. An **implicit cost** is the value of a resource used by the firm that could have been used elsewhere, even though no money changes hands. The owner's forgone salary and forgone interest on invested savings are examples.
> **Economic cost** is the opportunity cost of all resources used in production: explicit costs *plus* implicit costs.

An accountant ordinarily records payments made by the business. An economist also counts the income the owner sacrifices by using her own labor and capital in the firm.

> **KEY RESULT Accounting profit and economic profit**
>
> $$\text{Accounting profit} = \text{Total revenue} - \text{Explicit costs},$$
> $$\text{Economic profit} = \text{Total revenue} - \text{Explicit costs} - \text{Implicit costs}.$$
>
> Economic profit equals accounting profit *minus* implicit costs. Economic profit of zero means that the firm covers all costs, including the owner's opportunity costs. It is not the same as earning nothing.

Worked example: An engineer starts a business

An engineer gives up a \$120,000 annual salary and invests \$500,000 of savings that could earn 5% annually. The business earns revenue of \$1,000,000 and pays \$800,000 in explicit costs.

$$\text{Accounting profit} = \$1{,}000{,}000 - \$800{,}000 = \boxed{\$200{,}000},$$
$$\text{Implicit costs} = \$120{,}000 + 0.05(\$500{,}000) = \boxed{\$145{,}000},$$
$$\text{Economic profit} = \$200{,}000 - \$145{,}000 = \boxed{\$55{,}000}.$$

The business earns \$55,000 more than the owner's specified next-best alternatives. Positive accounting profit alone would not establish that conclusion. Over the long run, a persistently negative economic profit is a reason to redeploy resources; it does not automatically imply immediate shutdown when some costs cannot be avoided in the short run.

2. Production: the law of diminishing marginal productivity

DEFINITION Production function

A **production function** describes the maximum output a firm can produce from specified quantities of inputs, given its technology. In the **short run**, at least one input is fixed. With a fixed factory (capital $\bar{K}$) and variable labor L, write

$$Q = f(L, \bar{K}).$$

It shows how output changes as the firm adds workers to the same factory.

DEFINITION Marginal and average product

The **marginal product of labor (MP)** is the additional output obtained from an additional unit of labor. The **average product of labor (AP)** is output per unit of labor:

$$\text{MP} = \frac{\Delta Q}{\Delta L}, \qquad \text{AP} = \frac{Q}{L} \quad (L > 0).$$

With whole workers, MP measures the change from hiring the next worker. For a smooth production function, MP can also mean the slope at a particular level of labor.

KEY RESULT Law of diminishing marginal productivity

As more labor is added while the factory and other inputs are held fixed, the extra output produced by an additional worker *eventually falls*. MP need not fall from the first worker. Initially, specialization can make MP rise; later, workers have to share the same fixed equipment and space.

Worked example: Output from a fixed factory

Suppose production is $Q = 10L + 9L^2 - L^3$. The table uses whole workers; the MP column is the increase in output when that worker is hired.

Workers L	Output Q	Added output (MP)	AP = Q/L	What happens
0	0	—	—	No labor employed
1	18	18	18	The first worker starts production
2	48	30	24	MP rises
3	84	36	28	MP rises
4	120	36	30	MP stops rising
5	150	30	30	MP has begun to fall
6	168	18	28	Further diminishing MP
7	168	0	24	The seventh worker adds no output

The table shows a first phase of increasing MP, followed by diminishing MP. Eventually, the extra worker adds

nothing. Further additions could even reduce output if they obstruct production.

> **KEY RESULT Relationship between marginal product and average product**
>
> When MP > AP, average product rises. When MP < AP, average product falls. On the usual smooth, single-peaked curves, MP crosses AP *from above* at the maximum of AP.
>
> Reason: a new worker producing more than the current average raises output per worker. A worker producing less than the current average pulls it down.

The production diagrams. The next two figures show the continuous curves corresponding to the same function. The smooth MP is the slope of the total-product curve, so its exact peak differs slightly from the whole-worker MP column.

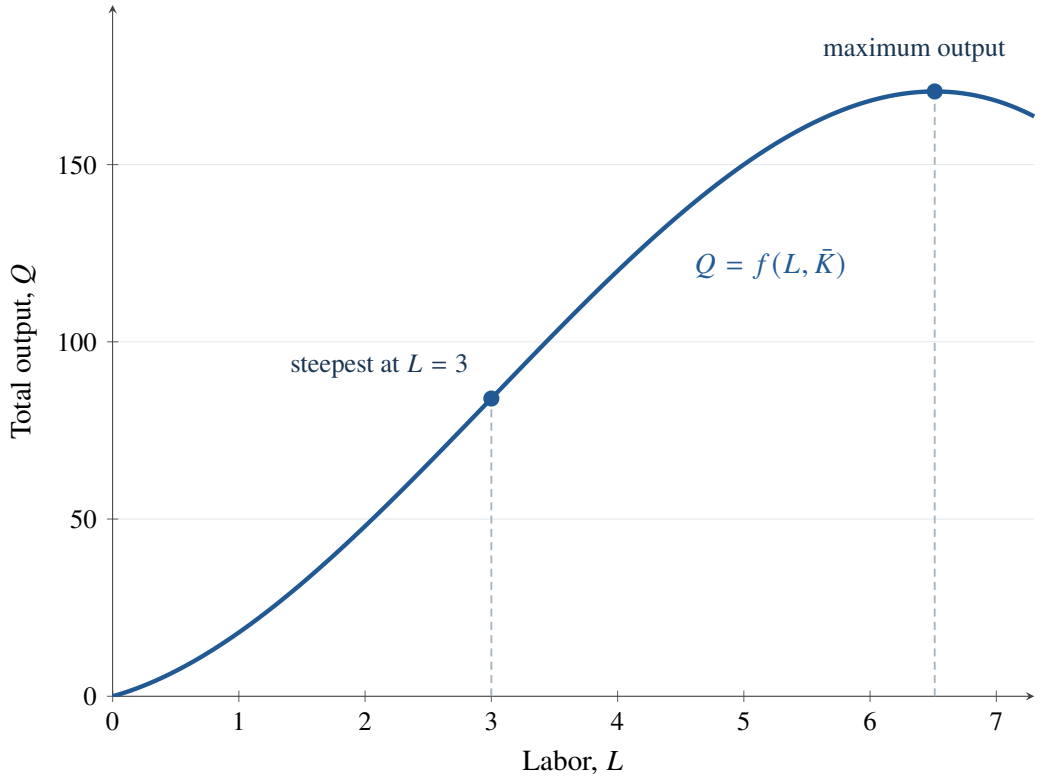


Figure 1. Total product first rises at an increasing rate, then at a decreasing rate. It reaches its maximum when the extra output from another small increase in labor has fallen to zero.

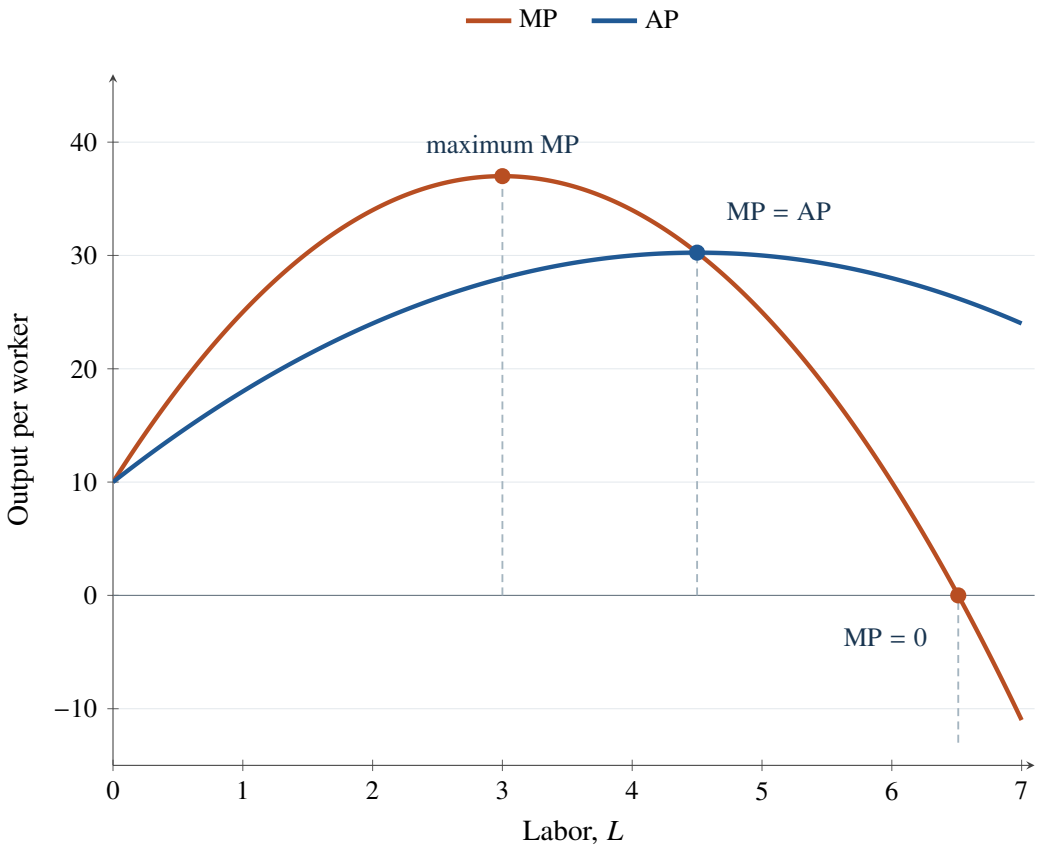


Figure 2. MP peaks before AP. When MP exceeds AP, AP rises; after MP crosses AP from above, AP falls. MP becomes zero when total output reaches its peak.

For this smooth example, MP is greatest at $L = 3$ (MP = 37), AP is greatest at $L = 4.5$ (MP = AP = 30.25), and total output is greatest at approximately $L = 6.51$ ($Q \approx 170.63$). The turning points need not coincide with whole-worker entries in the table.

3. Short-run costs: total, average, and marginal

DEFINITION Total, fixed, and variable cost

Fixed cost (FC) does not change when output changes in the short run, such as a factory lease. **Variable cost (VC)** changes with output, such as the cost of labor or materials. **Total cost (TC)** is their sum:

$$\mathrm{TC}(Q) = \mathrm{FC} + \mathrm{VC}(Q).$$

Fixed cost is incurred even when the firm produces nothing, provided the cost cannot be avoided during the period considered.

DEFINITION Average costs and marginal cost

Average cost means *cost per unit of output*. For positive output,

$$\text{AFC} = \frac{\text{FC}}{Q}, \qquad \text{AVC} = \frac{\text{VC}}{Q}, \qquad \text{ATC} = \frac{\text{TC}}{Q} = \text{AFC} + \text{AVC}.$$

Marginal cost (MC) is the additional cost of producing additional output:

$$\text{MC} = \frac{\Delta \text{TC}}{\Delta Q} = \frac{\Delta \text{VC}}{\Delta Q}.$$

The second equality follows because fixed cost does not change when output changes.

KEY RESULT Fixed cost versus average fixed cost

Total fixed cost stays constant as output rises. But **average fixed cost always falls** for $Q > 0$, because the same fixed sum is spread over more units.

$$\text{AFC} = \frac{\text{FC}}{Q} \downarrow \quad \text{as } Q \uparrow .$$

For example, a \$100 fixed cost is \$50 per unit at output 2, \$25 per unit at output 4, and \$12.50 per unit at output 8. Do not confuse constant FC with falling AFC.

Worked example: A firm's short-run costs

Continue with a cost function that is deliberately simple enough to calculate and draw:

$$\text{TC}(Q) = Q^3 - 9Q^2 + 30Q + 100.$$

Here FC = 100 and VC = $Q^3 - 9Q^2 + 30Q$. Dividing VC and TC by output gives AVC and ATC. The smooth marginal-cost curve measures the *slope* of TC at each output level:

$$\begin{aligned} \text{AVC}(Q) &= Q^2 - 9Q + 30, \\ \text{ATC}(Q) &= Q^2 - 9Q + 30 + \frac{100}{Q}, \\ \text{MC}(Q) &= 3Q^2 - 18Q + 30. \end{aligned}$$

Q	TC	FC	VC	AFC	AVC	ATC	MC at Q
0	100	100	0	—	—	—	30
1	122	100	22	100.00	22.00	122.00	15
2	132	100	32	50.00	16.00	66.00	6
3	136	100	36	33.33	12.00	45.33	**3**
4	140	100	40	25.00	10.00	35.00	6
5	150	100	50	20.00	10.00	30.00	15
6	172	100	72	16.67	12.00	28.67	30
7	212	100	112	14.29	16.00	30.29	51
8	276	100	176	12.50	22.00	34.50	78

Reading the MC column: These values are the slopes of the smooth cost function *at* each listed output, not the changes between adjacent rows. For instance, TC rises by $22 between outputs 5 and 6, while the smooth MC at output 6 is $30. Either measures a marginal change, but over different-sized intervals.

KEY RESULT The typical shapes of short-run cost curves

AFC falls continuously. In the usual short-run model, MC falls at first and then rises as diminishing marginal productivity sets in. AVC and ATC typically become U-shaped: they fall while marginal cost is below the relevant average and rise when marginal cost exceeds it. The minimum of MC occurs before the minima of AVC and ATC.

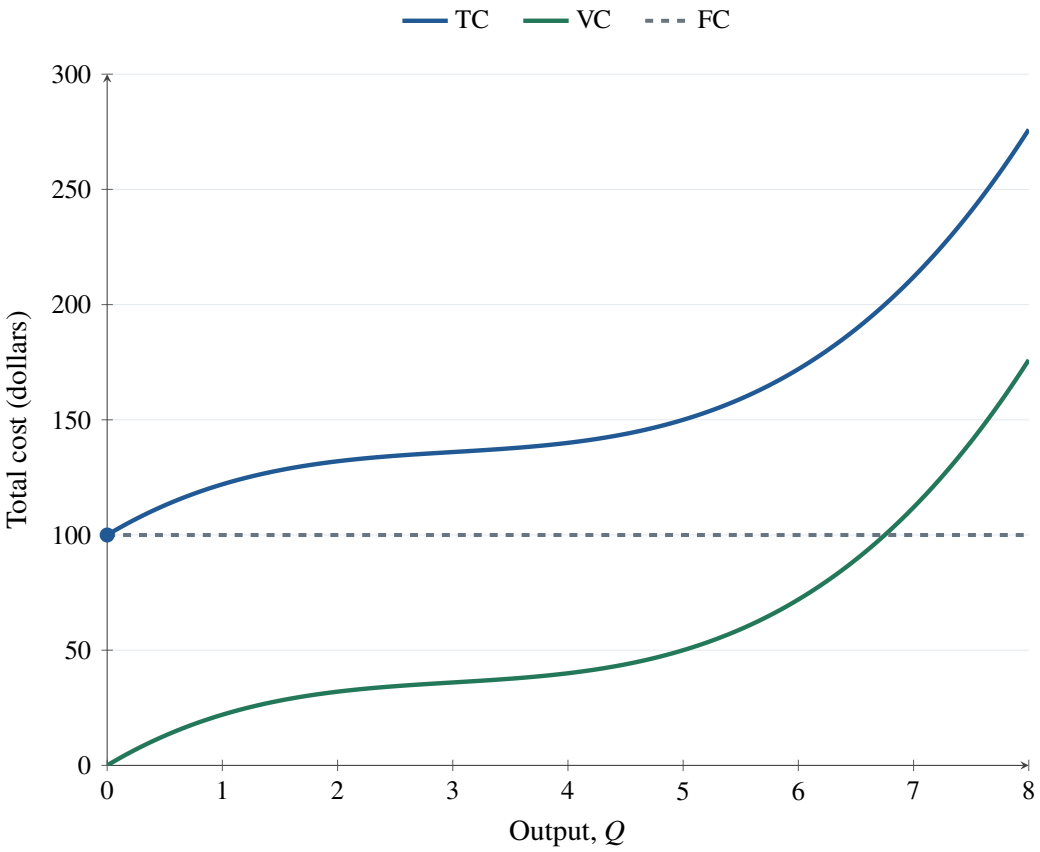


Figure 3. FC is horizontal. TC is always FC above VC. As output expands, the extra cost of another unit first falls and later rises, changing the steepness of TC and VC.

Read the graph. The distance between TC and VC stays at $100 at every output: that is fixed cost. The total-cost curve becomes increasingly steep at higher output because marginal cost is rising.

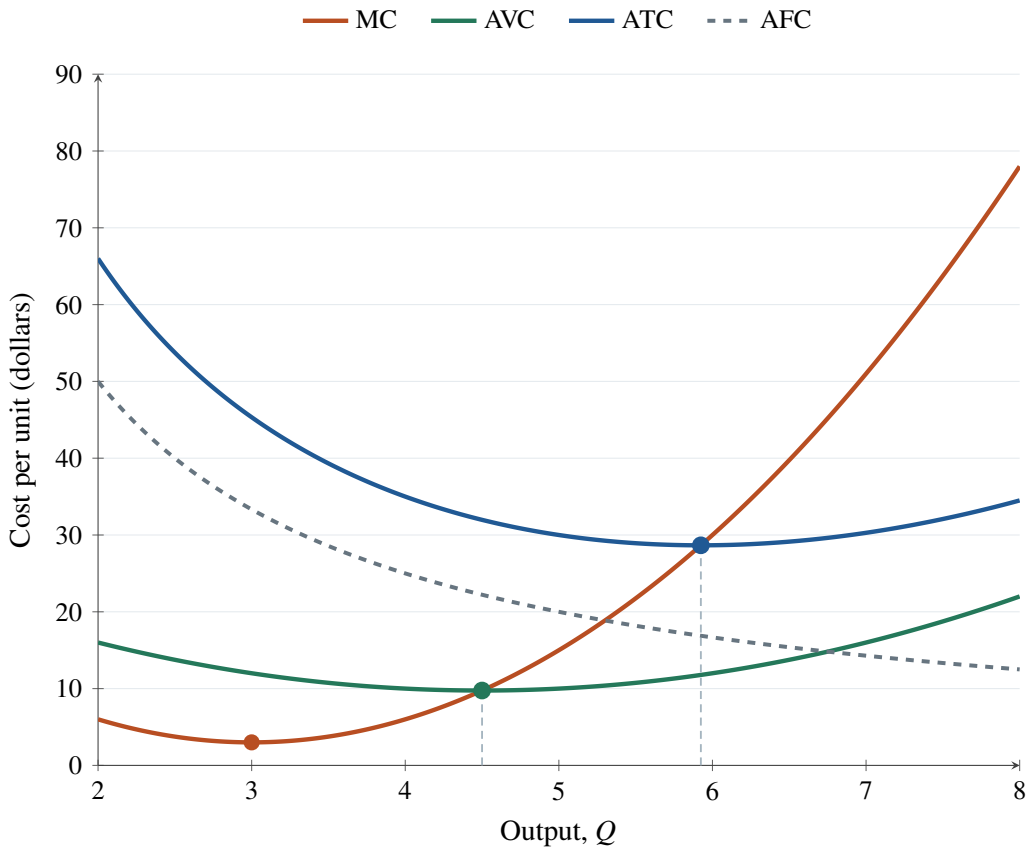


Figure 4. Short-run unit costs. AFC declines throughout. MC is lowest first (at $Q = 3$); MC meets AVC at its minimum ($Q = 4.5$) and ATC at its minimum ($Q \approx 5.92$). These intersections are marked by dots.

Read the graph. Follow the dashed AFC curve from left to right: it never turns upward. The U-shaped AVC and ATC curves move differently because variable costs and fixed costs affect their averages in different ways. The intersection rules are explained fully in Section 5.

4. The link between marginal cost and marginal product

Production determines how much labor is needed for a given output. Suppose labor is the only variable input and every unit of labor costs the same wage w. If the firm hires L workers, its variable cost is wL.

KEY RESULT Marginal product determines marginal cost

$$\boxed{\text{MC} = \frac{w}{\text{MP}}} \qquad \text{and} \qquad \boxed{\text{AVC} = \frac{w}{\text{AP}}}.$$

At a constant wage, a more productive additional worker makes each additional unit of output cheaper. Therefore,

$$\text{MP} \uparrow \Rightarrow \text{MC} \downarrow, \qquad \text{MP} \downarrow \Rightarrow \text{MC} \uparrow .$$

In the usual smooth example, **maximum MP corresponds to minimum MC**, and **maximum AP corresponds to minimum AVC**.

Why? Hiring one more worker adds wage cost w. If that worker produces MP additional units, the labor cost per additional unit is w/MP. Similarly, the total wage bill divided by total output is $wL/Q = w/\text{AP}$.

Worked example: A constant wage of \$36

Return to the production table in Section 2. Each additional worker costs \$36. The columns below use the *whole-worker* marginal product: for instance, the second worker adds 30 units, so the cost per unit added is $\$36/30 = \$1.20$.

L	Q	MP	AP	VC = $36L$	MC = 36/MP	AVC = 36/AP
1	18	18	18	36	2.00	2.00
2	48	30	24	72	1.20	1.50
3	84	36	28	108	**1.00**	1.29
4	120	36	30	144	**1.00**	**1.20**
5	150	30	30	180	1.20	**1.20**
6	168	18	28	216	2.00	1.29

As the worker's additional output rises from 18 to 36, marginal cost falls from \$2 to \$1. When the additional output subsequently falls, marginal cost rises. AVC is lowest where AP is highest.

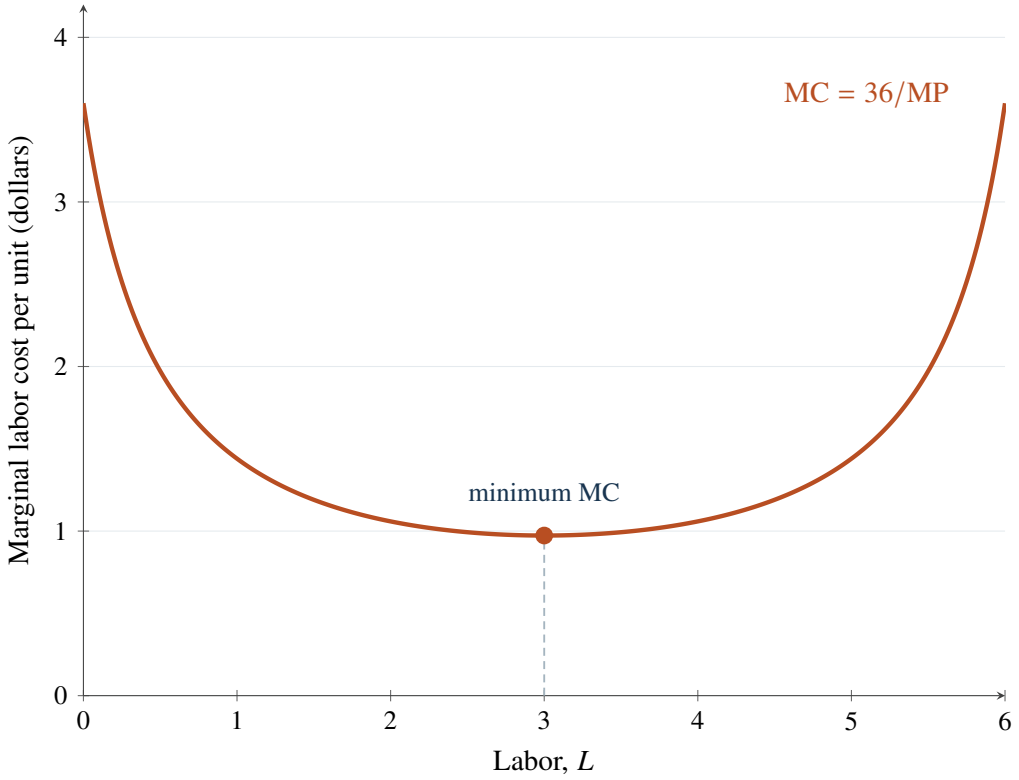


Figure 5. In the continuous version of the example, MP first rises and then falls as labor increases. With the wage fixed at \$36, MC moves in the opposite direction and reaches its minimum where MP is greatest ($L = 3$).

Read the graph. Compare the U-shaped MC curve here with the inverted-U MP curve in Figure 2. The lowest point of MC occurs at the highest point of MP. The graph uses the smooth MP curve; the whole-worker table uses discrete increases, so the numerical minima are not identical.

5. The link between marginal cost and average cost

This is the most important relationship for reading a firm's cost diagram. Think about an ordinary average: an additional observation below the existing average reduces it; an observation above it raises it. The same logic applies to average cost.

KEY RESULT The marginal–average rule

$$\text{MC} < \text{ATC} \Rightarrow \text{ATC falls}, \qquad \text{MC} < \text{AVC} \Rightarrow \text{AVC falls},$$

$$\text{MC} > \text{ATC} \Rightarrow \text{ATC rises}, \qquad \text{MC} > \text{AVC} \Rightarrow \text{AVC rises}.$$

On the usual smooth, U-shaped curves, **MC crosses ATC from below at the minimum of ATC**, and **MC crosses AVC from below at the minimum of AVC**:

$$\boxed{\text{MC} = \text{ATC at minimum ATC}}, \qquad \boxed{\text{MC} = \text{AVC at minimum AVC}}.$$

An equality by itself does not guarantee a minimum; it is the change from MC below the average to MC above it that gives the U-shaped minimum.

Worked example: Why the average falls or rises

At output 5 in the cost table, TC is \$150, so ATC is \$30. Increasing output from 5 to 6 adds \$22 to total cost. Because \$22 is *below* the old \$30 average, the new average falls to $\$172/6 \approx \$28.67$.

Increasing output again, from 6 to 7, adds \$40. Because \$40 is *above* the old \$28.67 average, the new average rises to $\$212/7 \approx \$30.29$. This is the entire marginal–average logic. No calculus is required.

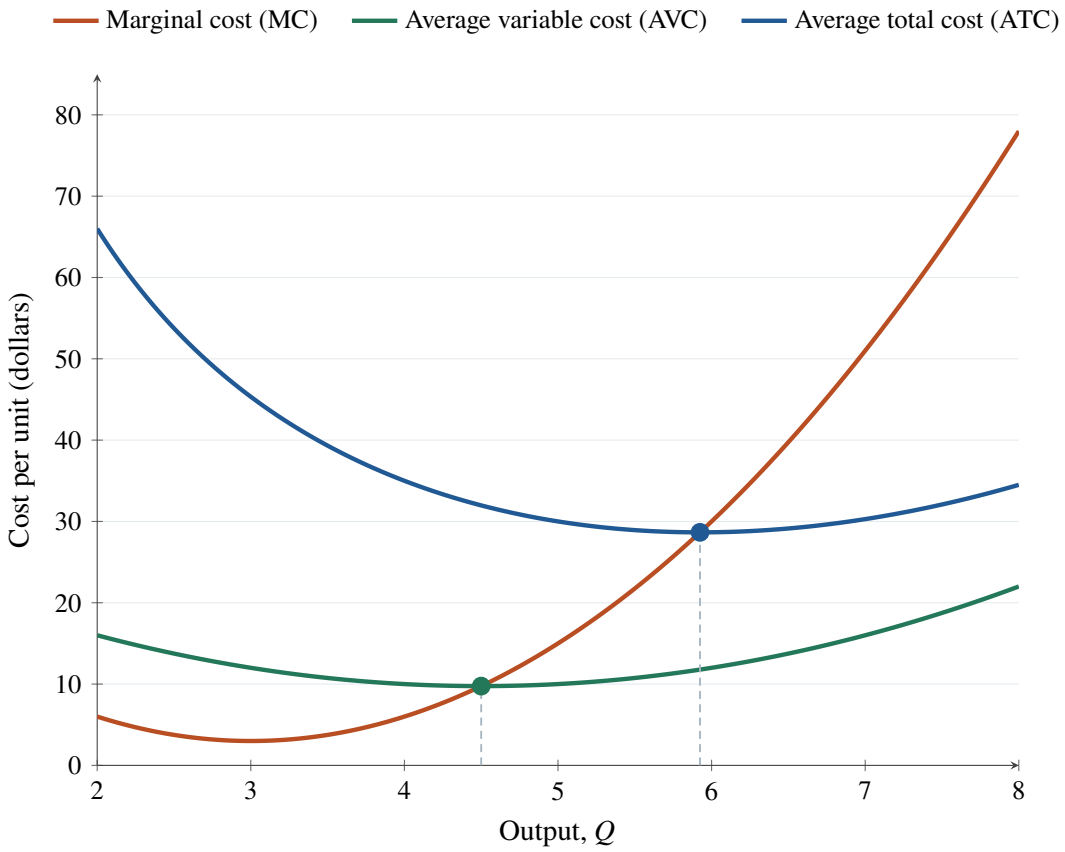


Figure 6. Marginal cost crosses AVC at $Q = 4.5$ and ATC at $Q \approx 5.92$, each from below. To the left of a crossing, MC is below that average and pulls it down; to the right, MC is above the average and pulls it up.

Read the graph. For AVC, locate the lower intersection ($Q = 4.5$). For ATC, locate the upper intersection ($Q \approx 5.92$). Do not confuse the two minima. They occur at different output levels.

> **KEY RESULT Why the ATC minimum occurs after the AVC minimum**
>
> ATC = AVC + AFC. At the output where AVC reaches its minimum, AFC is *still falling*. Therefore ATC is still falling at that point and reaches its own minimum at a higher output. In the example, minimum AVC is \$9.75 at $Q = 4.5$, while minimum ATC is about \$28.66 at $Q = 5.92$.

6. The long run: long-run average cost

DEFINITION Short run, long run, and long-run average cost

In the **short run**, at least one input is fixed, so the firm must work with its existing factory or plant. In the **long run**, all inputs, including plant size, can be changed.

Long-run average cost (LRAC) is the lowest possible average total cost of each output when the firm may choose all its inputs, including the plant:

$$\boxed{\text{LRAC}(Q) = \min_{k} \text{ATC}_k(Q)}.$$

Here k indexes feasible plant sizes. For every plant k and output Q, $\text{LRAC}(Q) \leq \text{ATC}_k(Q)$.

A small plant may be economical at a low output and expensive at a high output. A large plant may have the opposite advantage. The firm can choose a different plant as its scale changes.

KEY RESULT LRAC is the lower envelope of short-run ATC curves

Each plant has its own short-run average-cost curve. At every output, LRAC takes the *lowest* available cost among the feasible plants. The resulting curve is called the **lower envelope** of the short-run ATC curves. Where a particular plant is optimal, its short-run ATC touches LRAC. The optimal plant can change as output changes; the firm is not stuck with one short-run curve in the long run.

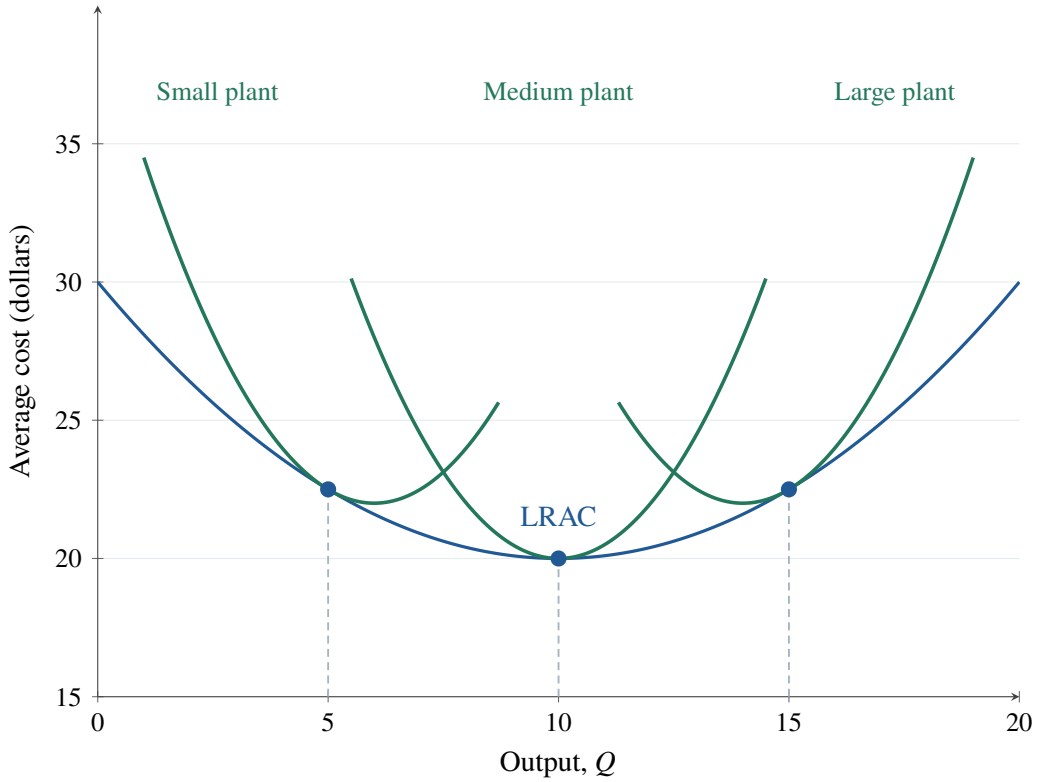


Figure 7. The blue LRAC curve is the lowest feasible average cost at each output. Green curves show three selected plant sizes: a small plant may be cheapest at low output, and a larger plant at higher output. The blue envelope reaches its minimum at $Q = 10$.

Illustrative curves. Here $\text{LRAC}(Q) = 30 - 2Q + 0.1Q^2$ and $\text{ATC}_s(Q) = \text{LRAC}(Q) + 0.4(Q - s)^2$, where s denotes the output suited to that plant. The figure draws only $s = 5, 10, 15$, but plants for intermediate values of s are also feasible. Minimizing across *all* s gives the blue envelope. If only the three plotted plants existed, the actual envelope would be higher between their contact points.

7. Economies and diseconomies of scale

DEFINITION Economies and diseconomies of scale

Economies of scale occur over a range of outputs when long-run average cost *falls* as output rises. **Diseconomies of scale** occur over a range when long-run average cost *rises*. Over a range with **constant average cost**, LRAC is flat as output changes.

The **minimum efficient scale (MES)** is the smallest output at which LRAC reaches its lowest level. In the illustrated U-shaped example, MES is $Q = 10$.

KEY RESULT Read the long-run average-cost curve

- **Falling LRAC:** economies of scale, as the firm lowers cost per unit by expanding.
- **Flat LRAC over an interval:** constant average costs over that interval.
- **Rising LRAC:** diseconomies of scale, as further expansion raises cost per unit.

A curve with a single minimum has zero slope at that one point, but *that alone* does not mean there is a range of constant returns to scale.

Why LRAC may fall. A larger firm may specialize its labor force more effectively or spread the cost of indivisible equipment over more units. **Why LRAC may rise.** A very large operation may face increasing costs of communication, supervision, and coordination.

Connection with returns to scale in production. Returns to scale ask what happens to output when *all* inputs are increased in the same proportion. If doubling an initially cost-minimizing input bundle more than doubles output, while input prices remain unchanged, the doubled bundle produces more than twice the original output at twice the cost; its average cost is lower. Input proportions may also change when the firm minimizes costs, so one cannot infer diseconomies of scale simply because a particular doubled input bundle produces less than twice the original output.

KEY RESULT Do not confuse the short run with the long run

Diminishing marginal product is a short-run idea: add more of a variable input while holding another input fixed. It explains why short-run MC eventually rises when the wage is fixed.

Diseconomies of scale is a long-run cost idea: LRAC increases as output expands and the firm is free to adjust *all* inputs. Rising short-run MC is not, by itself, evidence of diseconomies of scale.

The essential cost relationships. Fixed cost stays fixed; average fixed cost falls. Diminishing marginal product causes marginal cost to rise in the usual short-run model. Marginal cost below an average pulls that average down, and marginal cost above it pushes that average up. Hence, on the standard U-shaped curves, MC crosses AVC and ATC at their separate minima, from below. In the long run the firm can change every input and chooses the lowest feasible average cost at each output.